

TPC-386: A Count Holdout for the Finite $c = 1$ Bandwidth Proxy

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September 4, 2026

Abstract

The preceding TPC-385 experiment transferred a finite high-bandwidth phase between fresh origins while keeping the window count fixed. We now freeze that reference and change the count. Three coordinate-disjoint windows of length 512 define a calibration-only pooled geometry, and two windows of length 1024 are evaluated as a response-blind holdout. Both a fixed three-block band and the full relative band are recorded. In the all-plus law at $Q = 8192$, the holdout-to-calibration spectral ratios range from 1.0652 to 1.1295, while the inherited finite spectral diagnostic 0.64 fails on all sixteen all-plus holdout rows. The failure already appears in the fixed band, so it is not caused solely by remote block pairs. The result is a finite count-transfer observation and a scoped obstruction to promoting the old cap to a count-uniform statement. It proves no growing operator bound and makes no arithmetic claim about twin primes.

1 Question and claim boundary

We work in one fixed finite dynamical-system family. The question is whether the origin transfer observed in TPC-385 survives a predeclared change from $N = 512$ to $N = 1024$, and whether its diagnostic spectral cap can be reused. The distinction matters: a finite origin holdout and a bound uniform in the window count are different claims. Throughout this paper,

ARITHMETIC_ADVANCE=NO, FIXED_POWER_CREDIT=0, FULL_GATE_B=OPEN.

The official Session evaluator files are not present in this checkout; the local Bridge-B checker is therefore treated as fail-closed repository evidence rather than an official route verdict.

2 Finite proxy and frozen protocol

For $p \in (Q, 2Q]$ and u, v in a consecutive integer interval I , set

$$K_p(u, v) = p(p/Q)^2 \frac{66^2}{66^2 + (u - v)^2} \left(\mathbf{1}_{p|u-v} - \frac{1}{p-1} \right) \\ \cdot \mathbf{1}_{u \neq v} \mathbf{1}_{p \nmid u} \mathbf{1}_{p \nmid v}.$$

For each of the four declared sign laws σ , let $K_\sigma = \sum_{p \in (Q, 2Q]} \sigma(p) K_p$ and define the diagonal geometry

$$G(u) = \sum_{p \in (Q, 2Q]} \sum_{v \in I} K_p(u, v)^2.$$

The local matrix divides $K_\sigma(u, v)$ by $\sqrt{G(u)G(v)}$. The pooled matrix divides by the mean of G over the three calibration windows only. Blocks have length 128.

The candidate origins are $a_j = 2200001 + 401j$ for $0 \leq j < 41$; indices 0, 10, 20, 30, 40 are selected before readout. Indices 0, 10, 20 have $N = 512$ and are calibration, while indices 30, 40 have $N = 1024$ and are holdout. The fixed band retains block pairs at distance at most three. The full-relative band retains all block pairs at the relevant count. We use $Q = 2048, 8192$, exponent one, $\beta = 2$, and the laws `all_plus`, `alternating_index`, `mod4_character`, and `half_split`.

Table 1: All-plus $Q = 8192$ count transfer. The parent reference is the locked TPC-385 $N = 512$ value; “ratio” compares holdout with calibration.

band	normalization	calibration	holdout	ratio	$\log_2$ ratio
fixed $c = 3$	local	0.620802645	0.661288692	1.065216	0.091146
fixed $c = 3$	pooled	0.638885619	0.709942760	1.111220	0.152145
full relative	local	0.620802645	0.674100952	1.085854	0.118830
full relative	pooled	0.638885619	0.721585871	1.129445	0.175613

For a masked symmetric matrix B we record $\|B\|_2$, its maximum absolute row mass (Schur diagnostic), its Frobenius norm, and its symmetry error. The count ratio is the $N = 1024$ holdout mean divided by the $N = 512$ calibration mean within one fixed cell. A 20% ratio envelope is an explicitly declared finite audit threshold, not a theorem.

3 Certificate and exact anchor

The producer writes a canonical JSON certificate with 160 rows and 32 cells. The certificate locks the TPC-385 producer and certificate hashes, records the response-blind selection protocol, and includes an independent replay which sums the prime shell in reverse order. A second script applies 25 structural mutations; all are rejected by the certificate validator.

At $Q = 8$, the exact interval $[2200001, 2200014)$ has shell $\{11, 13\}$. Rational arithmetic verifies positive diagonal geometry and symmetry of all four signed matrices on this anchor. This exact check validates the finite construction, not its large- N behavior.

4 Results

The complete panel has no Schur-cap failure. There are 16 spectral-cap failures, exactly the all-plus rows at $N = 1024$ for the two modes, two normalizations, and two values of Q . The calibration and holdout one-percent stability counts are respectively 20/32 and 28/32.

All four ratios lie inside the declared 20% finite audit envelope, but that observation does not establish count uniformity. Relative to the locked parent values 0.6207997105 (local) and 0.6388876036 (pooled), the four holdout forecast errors are 6.5221%, 11.1217%, 8.5859%, and 12.9441%, in the table order. The fixed-band local holdout value already exceeds 0.64, and the full band is only modestly larger. Thus the old cap cannot be transferred to this count change even in a band that excludes the most remote block pairs.

The signed laws remain controls rather than a uniformity certificate. For example, in the full-relative, $Q = 8192$ cell their holdout means are 0.000581599, 0.004598884, and 0.218454930 for the alternating, mod-4, and half-split laws, respectively. Their different scales and spreads reinforce the need to keep the all-plus transfer claim scoped.

5 Interpretation and next step

The strongest positive result is a reproducible, origin-disjoint count transfer inside a broad finite envelope. The strongest obstruction is the failure of the inherited 0.64 spectral diagnostic on every all-plus $N = 1024$ row, including the fixed-band mode. The reusable structure is a locked parent forecast, calibration-only pooled normalization, explicit count roles, and reverse-order plus mutation checks.

The next minimal question is whether the observed increase is a smooth count ladder effect or an artifact of the endpoint normalization. The project therefore emits

ROUND2_CLUE=TEST_C1_COUNT_LADDER_RENORMALIZATION.

Any resulting statement must remain finite until a genuine growing- N argument is supplied.

Reproduction

The source, certificate, proof package, and checkers are in `papers/tpc-386-c1-count-holdout-bandwidth/`. The release runs the producer and independent checker in ordinary and optimized Python modes, the 25-mutation stress test, and the local Bridge-B checker. The generated files `paper/main.pdf` and `paper/paper.pdf` are byte-identical.